

Vector Spaces, subspaces, dimension

Real and Complex Vectors

An n -dimensional real vector is an n -tuple $(x_1, \dots, x_n)$ where the x_i are real numbers. If the x_i are complex instead of real, we have an n -dimensional complex vector.

We write $\mathbf{R}^n$ and $\mathbf{C}^n$ for the sets of n -dimensional real and complex vectors respectively.

One can visualize an element $x = (x_1, \dots, x_n)$ of $\mathbf{R}^n$ as an “arrow” with its base at the origin and its endpoint at the point $(x_1, \dots, x_n)$ in n -dimensional Euclidean space.

Elements of $\mathbf{R}$ or $\mathbf{C}$ are often called *scalars*.

Operations

The set $\mathbf{R}^n$ allows for:

Addition of vectors. If

$$x = (x_1, \dots, x_n) \in \mathbf{R}^n \text{ (or } \mathbf{C}^n)$$

and

$$y = (y_1, \dots, y_n) \in \mathbf{R}^n \text{ (or } \mathbf{C}^n)$$

then

$$x + y = (x_1 + y_1, \dots, x_n + y_n)$$

is in $\mathbf{R}^n$ (or $\mathbf{C}^n$).

Multiplication of a vector by a scalar. If

$$a \in \mathbf{R} \text{ and } y = (y_1, \dots, y_n) \in \mathbf{R}^n$$

then $ay = (ay_1, \dots, ay_n)$. Similarly if $a \in \mathbf{C}$ and $y \in \mathbf{C}^n$.

Axioms

These operations satisfy a list of axioms. For simplicity let's work with $\mathbf{R}$ and $\mathbf{R}^n$ but the same things are true for $\mathbf{R}^n$ and $\mathbf{C}^n$.

1. Vector addition is commutative, associative, and if $e = (0, \dots, 0)$ then $x + e = e + x = x$.
2. Scalar multiplication distributes over addition so that $(a + b)x = ax + bx$ and $a(x + y) = ax + ay$.
3. Scalar multiplication is "associative" in the sense that $a(bx) = (ab)x$.
4. $1x = x$ and $0x = e$.

Real (and Complex) Vector spaces

Definition: A *vector space* over $\mathbf{R}$ is a set V together with two operations: - Addition. Given x, y in V , there is an element $x + y \in V$. - Scalar Multiplication. Given $a \in \mathbf{R}$ and $x \in V$ there is an element $ax \in V$.

The two operations must satisfy the axioms listed above. In particular there must be an element $e \in V$ so that $x + e = e + x = x$ for all $x \in V$.

A vector space over $\mathbf{C}$ is the same except that scalar multiplication is defined for $a \in \mathbf{C}$, not just $\mathbf{R}$.

The spaces $\mathbf{R}^n$ and $\mathbf{C}^n$ above are examples of real and complex vector spaces respectively.

Subspaces

Definition: A subspace W of a vector space V (for either $\mathbf{R}$ or $\mathbf{C}$) is a subset of V that is a vector space under the inherited operations from V .

Example: Consider the set W of vectors $(x_1, x_2, 0)$ inside $\mathbf{R}^3$. This is a copy of $\mathbf{R}^2$ sitting inside $\mathbf{R}^3$ so it's a subspace.

Example: Let W be the set of vectors in $\mathbf{R}^n$ that satisfy $x_1 + x_2 + \dots + x_n = 0$. This is a subspace. Check: - If $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$ are in W , then $u + v$ is in W because

$$(u_1 + v_1) + \dots + (u_n + v_n) = (u_1 + \dots + u_n) + (v_1 + \dots + v_n) = 0$$

Similarly $au_1 + au_2 + \dots + au_n = a(u_1 + \dots + u_n) = 0$ so au is in W .

Finally $e = (0, \dots, 0)$ is in W .

The other axioms hold for elements of W since all elements of W are also elements of V .

Example Let W be the set of vectors in $\mathbf{R}^n$ that satisfy $x_1 + \cdots + x_n = 1$. This is **NOT** a subspace.

Proposition: A subset W of V is a subspace provided that it contains the element e and whenever $u, v \in W$ it follows that $u + v \in W$ and $au \in W$ for all $a \in \mathbf{R}$.

You can replace $\mathbf{R}$ by $\mathbf{C}$ in the above to obtain the same results for complex vector spaces.

Four Crucial Definitions

Span of a set of vectors

Definition: Let V be a vector space and let $S = \{v_1, \dots, v_k\}$ be a set of vectors in V . The **span** of S is defined as either of the following equivalent things:

1. The set of all vectors $\sum_{i=1}^k a_i v_i$ as $(a_1, \dots, a_k)$ runs over $\mathbf{R}^k$.
2. The intersection of all subspaces of V that contain S .

The set H in 1 is a subspace that contains S . If W is the intersection of all subspaces containing S , then $W \subset H$ since H is among the sets being intersected. On the other hand, since S is contained in W , and W is a subspace, the elements in H are also in W . So $W = H$.

Linear Independence

Definition: A set of vectors $v_1, \dots, v_n$ in a vector space V is *linearly independent* if the only solution $(a_1, \dots, a_n)$ to the equation

$$\sum_{i=1}^n a_i v_i = 0$$

is $(0, \dots, 0)$.

Equivalently:

- a set $S = \{v_1, \dots, v_n\}$ is linearly independent if the span of any proper subset of S is a proper subspace of the span of S .

Let $H = \text{Span}(S)$.

Suppose

$$\sum a_i v_i = 0$$

and that $a_j \neq 0$. Then you can “solve” for v_j in terms of the other vectors:

$$v_j = -\sum_{i \neq j} a_i / a_j v_i$$

so v_j is in the span of the remaining v_i . Thus the span of the set $S \setminus \{v_j\}$ is the same as H .

Conversely, if there is a proper subset U of S whose span equals H , choose $v_i \notin U$. Since v_i is in the span of U , you can write

$$v_i = \sum_{v_j \in U} b_j v_j$$

and therefore

$$-v_i + \sum b_j v_j = 0$$

and so S is not linearly independent.

- a set $S = \{v_1, \dots, v_n\}$ is not linearly independent (is *dependent*) if and only if there is some $1 \leq j \leq n$ such that

$$v_j = \sum_{\substack{i=1 \\ i \neq j}}^n a_i v_i$$

This is implicit in the argument above.

Basis

A set S of vectors in a vector space V is called a *basis* of V if $\text{Span}(S) = V$ and S is linearly independent.

Dimension

Theorem: Every vector space has a basis.

Proof: This requires Zorn's Lemma. If $S_1 \subset S_2 \subset \dots$ is an ascending sequence of linearly independent sets, their union is also linearly independent (because any linear relation would only involve finitely many vectors so would hold at some finite level of the chain). Therefore the linearly independent subsets of V form a partially ordered set in which every chain has an upper bound (namely its union). By Zorn's Lemma, it follows that there is a maximal linearly independent set B in V .

Now choose any $v \in V$. If v is not in the span of B , then $B \cup \{v\}$ would be linearly independent, contradicting maximality of B .

Hence B spans V and is therefore a basis.

Theorem: Any two bases of a vector space V have the same cardinality.

Proof: Suppose that X and Y are bases of the same vector space V . If X is finite, this is the "same number of elements" fact established later by the exchange argument, so assume X is infinite.

For each $x \in X$, since Y spans V , x is a *finite* linear combination of elements of Y ; let $Y_x \subset Y$ be the (finite) set of elements of Y that appear with nonzero coefficient in that combination.

Claim: $Y = \bigcup_{x \in X} Y_x$. If some $y_0 \in Y$ belonged to no Y_x , then, since X spans V , y_0 is a finite linear combination of elements of X ; substituting each of those x 's by its expansion in terms of Y writes y_0 as a combination of elements of $Y \setminus \{y_0\}$ (none of the terms involve y_0 , since $y_0 \notin Y_x$ for any x used). This contradicts the linear independence of Y .

A union of a family of finite sets Y_x indexed by an infinite set X has cardinality at most that of X (see Hessenberg's Theorem). Applying this to $Y = \bigcup_{x \in X} Y_x$ gives $|Y| \leq |X|$.

X and Y play symmetric roles (both are bases), so the same argument with X and Y exchanged gives $|X| \leq |Y|$. By the Cantor-Bernstein theorem, $|X| = |Y|$.

We will establish the fact that any two finite bases of V have the same number of elements a little later.

Definition: The dimension of V is the cardinality of any basis of V .

If V has a finite basis, it is said to be finite-dimensional; otherwise it is infinite-dimensional.